

Blake Yang

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🐙 GitHub: github.com/sand050965 | 🧑 Portfolio: https://portfolio-1d8v-sand050965s-projects.vercel.app

EDUCATION

Northeastern University

M.S. Computer Science

Sep. 2025 – May. 2027

Seattle, WA

- Courses: Distributed Systems, Machine Learning, Database, Algorithms and Data Structures, Object-Oriented Programming

EXPERIENCE

Software Development Engineer Intern

Amazon Web Services (AWS)

May. 2026 – Aug. 2026

Seattle, WA

- Built **EventBridge**-triggered AWS **Lambda** data pipelines in **Java** and **TypeScript** that hydrated **ElastiCache** for **Redis** with **850K+** records from **API Gateway** and **S3** sources, using **S3** as a fallback store.
- Extended an existing **Lambda**-based monitoring service in **Java** with a filtering layer backed by **Redis** lookups, suppressing false-positive alerts across **21M+** accounts and cutting on-call noise by **90%**.
- Augmented the monitoring service in **Java** with tier-aware data prioritization, surfacing the cohort generating **98%** of business-critical revenue and accelerating critical-incident response by **43%**.

Software Engineer

Tryandco

Jun. 2023 – Dec. 2024

Taipei, Taiwan

- Developed back-end and web development solutions for **25+** projects using **PHP (Laravel)**, **MySQL**, and **Vue.js** to build full-stack CMS platforms, resulting in a **45%** improvement in clients' workflow efficiency.
- Migrated **20+ EC2** workloads to **AWS Lambda** and leveraged **unit-testing** to improve system scalability and reduce infrastructure costs, achieving **\$2,000** in annual cost savings.
- Implemented **40+** serverless workflows using **AWS Lambda**, integrating **REST APIs** and testing with **Postman**, supporting up to **1,000** concurrent requests without downtime and improving **45%** operational efficiency.
- Leveraged **load balancing** with **AWS ALB** and **caching** with **ElastiCache** for **Redis** to handle high traffic requests efficiently, improving system throughput by up to **40%** and reducing database load by **37%**.

Backend Engineer

Cathay Life Insurance

Apr. 2021 – Sep. 2022

Taipei, Taiwan

- Utilized **Java (Spring Boot)** and **MySQL** to build **REST APIs** for the internal system with integration testing, enhancing system performance by **30%**.
- Programmed and debugged Java-based data migration batch jobs in an Agile environment, reducing transfer time for **10 million+** records of critical business data by **55%**.
- Refactored legacy **Java** code into modular, unit-testable classes and redesigned the **MySQL** database schema, achieving **65%** higher query performance for new modules and enabling robust business intelligence reporting.

PROJECTS

Chatflow (Distributed Real-Time Chat System)

- Architected a distributed real-time chat system using **Java (Spring Boot)**, **PostgreSQL**, **RabbitMQ**, and **Redis**, deploying 4 horizontally scaled **WebSocket** servers on **AWS EC2** behind an ALB with async message queuing and cross-server pub-sub — stress-tested with **JMeter** up to **1M** messages across **256** concurrent threads.

PipelineForge (Distributed CI/CD Pipeline Execution System)

- Built a distributed CI/CD pipeline execution system using **Java (Spring Boot)**, **RabbitMQ**, **PostgreSQL**, and **Docker** on **Kubernetes** with **Helm**, integrating **Grafana** and **Loki** for observability, supporting **10** concurrent executions via async **RabbitMQ** consumers.

Mini Room (Full-Stack Real-Time Conferencing System)

- Crafted a real-time video conferencing system using **Python (Flask)** and **MySQL** and deployed a **Dockerized** backend on **AWS**, reducing API response time by **40%** through **caching** and integrating **third-party APIs** for payment.

Hand Tracking Canvas (Gesture-Controlled Drawing App with Computer Vision)

- Developed a **Python** desktop drawing application enabling touchless drawing via real-time hand gesture recognition using **OpenCV** and **MediaPipe**, improving FPS by **60%** through frame skipping for smooth interaction.

TECHNICAL SKILLS

Languages: Java, Python, TypeScript, JavaScript, C++, Golang, PHP, Node.js, HTML5, CSS, SQL

Frameworks: Spring Boot, Flask, Laravel, Express.js, React.js, Vue.js

Database (SQL & NoSQL): MySQL, PostgreSQL, MongoDB, Redis

Cloud Services: AWS – EC2, Lambda, RDS, DynamoDB, ALB, ElastiCache, SQS, S3, EventBridge, API Gateway, CloudWatch, CloudFront, CodeDeploy, VPC

Developer Tools: Git, GitHub, Docker, Kubernetes, Helm, RabbitMQ, Grafana, Loki, Nginx, Linux, MacOS, REST API, Postman, CI/CD, DevOps, Agile, Unit Testing, Integration Testing